

Preparation List – QUESTION 1

No.	Reagents/Apparatus/Chemicals	Quantity per pax
1	Mung bean seeds (<i>Vigna radiata</i>) previously soaked for 24 hours in tap water	20
2	Soda lime pellets in a capped container labelled: “ Soda lime, harmful, corrosive ”	10 g
3	Respirometer	1
4	Stopwatch	1
5	30 cm ruler (mm)	1
6	Colored solution in a small container labelled “ dye ” • Water with red or blue food dye added	5 cm ³
7	Marble-size ball of Plasticine®	1
8	Small ball of cotton wool (50-cent coin size)	1
9	Blunt forcep	1
10	Access to mass balance (0.01g accuracy)	8 per lab
11	Safety glasses	1
12	Latex gloves	1 pair
13	Spatula	1
14	Paper towels	5
15	Glass rod	1

Instruction for preparation

Mung beans

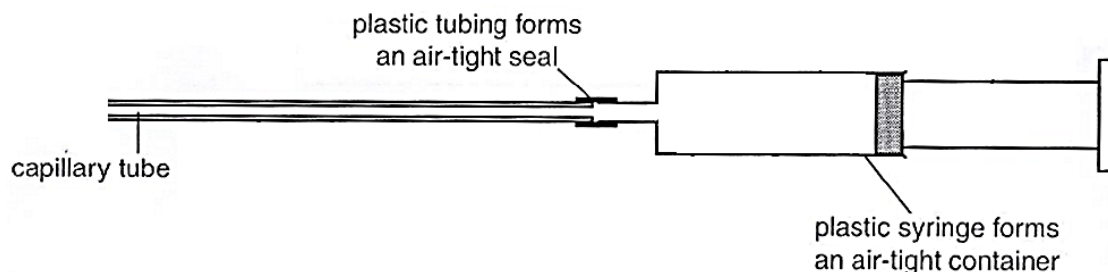
- Container used to be large enough for the expansion of the beans
- Water height just enough to cover the beans

Soda lime

- Pellets of soda lime can be packed in advance in air-tight plastic containers

Respirometer

- 20-cm³ syringe
- 20-cm length capillary tube of 1mm bore
- Plastic tubing



- Syringes used should be air-tight plunger that does not leak when tested by immersing in water.
- Tubing must be tight enough to prevent air leakage. This can also be tested by immersing the joint in water.

Preparation List – QUESTION 2

No.	Reagents/Apparatus/Chemicals	Quantity per pax
1	In a capped container, labeled K , 10% urease solution	25 cm ³
2	In a capped container, labeled U , 10% urea solution	25 cm ³
3	In a beaker, labeled W , distilled water	100 cm ³
4	Red litmus paper (6 x 1 cm)	4
5	Test-tube rack, suitable for holding 6 test-tubes	1
6	Test-tubes	6
7	Small beakers to hold up to 50 cm ³	6
8	Paper towels	8
9	10 cm ³ syringe	2
10	5 cm ³ syringe	2
11	White tile	1
12	Glass rod	1
13	Blunt forceps	1
14	Scissors to cut litmus paper	1
15	Ruler (mm)	1
16	Container, labeled ' waste '	1
17	Stopwatch	1
18	Marker	1
19	Safety glasses	1
20	Access to sink and tap water	-

Preparation of solutions

Solution **E** – 10% urease solution at room temperature

- Dissolve 10 g of urease active meal or three crushed tablets of urease (according to manufacturer's instructions) in a beaker with 50 cm³ of distilled water
- Make up to 100 cm³ with distilled water
- Mix well
- The solution may remain cloudy.

Solution **U** – 10% urea solution at room temperature

- Add 10 g of urea to 80 cm³ of distilled water in a beaker
- Make up to 100 cm³ with distilled water
- Mix well

Testing the activity of urease

- Before the practical, put a small piece of red litmus paper into a dry test-tube. Add 2 cm³ of **E** then 2 cm³ of **U** and start timing. Record time taken for the litmus paper to start turning blue.
- If this is longer than 5 minutes, increase the concentration of urea to 15%.
- It is not necessary to inform students that the concentration is different from that given in the question paper.